THE BUTTERFLY EFFECT OF BIOTECH

How innovation that is centered around improving and saving life maximization faces a multifaceted paradox and macro scarcity due to inefficient markets

INTRODUCTION AND MOTIVATION: THE VALUE OF HEALTHCARE

- If I offered, you all 1 billion dollars today but the caveat was that you would not wake up tomorrow…would you accept?
- Innovation in healthcare can be attributed to buying time
- Innovation in healthcare is maximized when macroeconomic conditions lend themselves towards this innovation- given the sector’s fragility towards business cycles and emotional sentiment thereof

INTRODUCTION AND MOTIVATION: THE PROBLEM

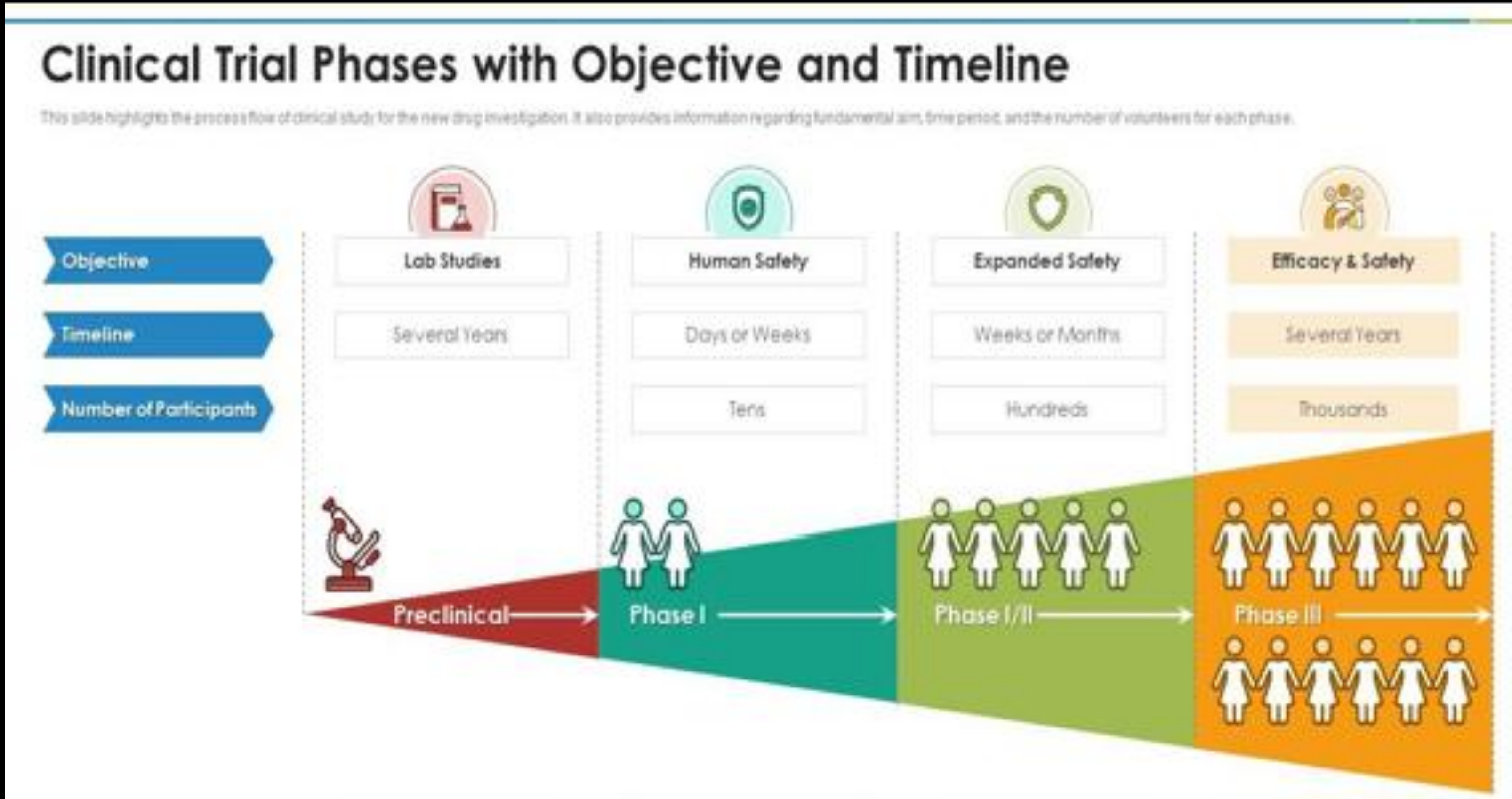
- In 2021 we saw expansionary monetary policy (low interest rates and high inflation) in the U.S. and **tech investment surged 104%,** while biotech faced a **52% funding drop**
- This details **"the flight to quality"** experienced during **volatile monetary policy (2021-2025)**
- Where market sentiment and narrative driven investments **"investor fomo"** take over and cause opportunity costs in humanities most important sector

INTRODUCTION AND MOTIVATION: WHY THE FUNDING DROP? THE EQUITY PARADOX

- **Micro and small cap biotech** companies were responsible for **56% of New Molecular Entities in 2023**
- These same companies face a paradox in which their value is reliant on their **potential cash flows (clinical stage assets),** yet their potential cash flows are reliant on **equity value (for their cash runways)**
- The need for **niche sector knowledge** in the sector leads to **inefficient displacements of capital during high inflation expansionary policy**
- The need of **concentrated investment from entities: (VC, PE, and banks alike)** exaggerates this problem during the mentioned overly expansionary policy **("once burned, twice shy")**
- Clinical timelines are long while investors have adapted a "cash in cash out" perspective for tech and **Large cap stocks (2021-2025)**

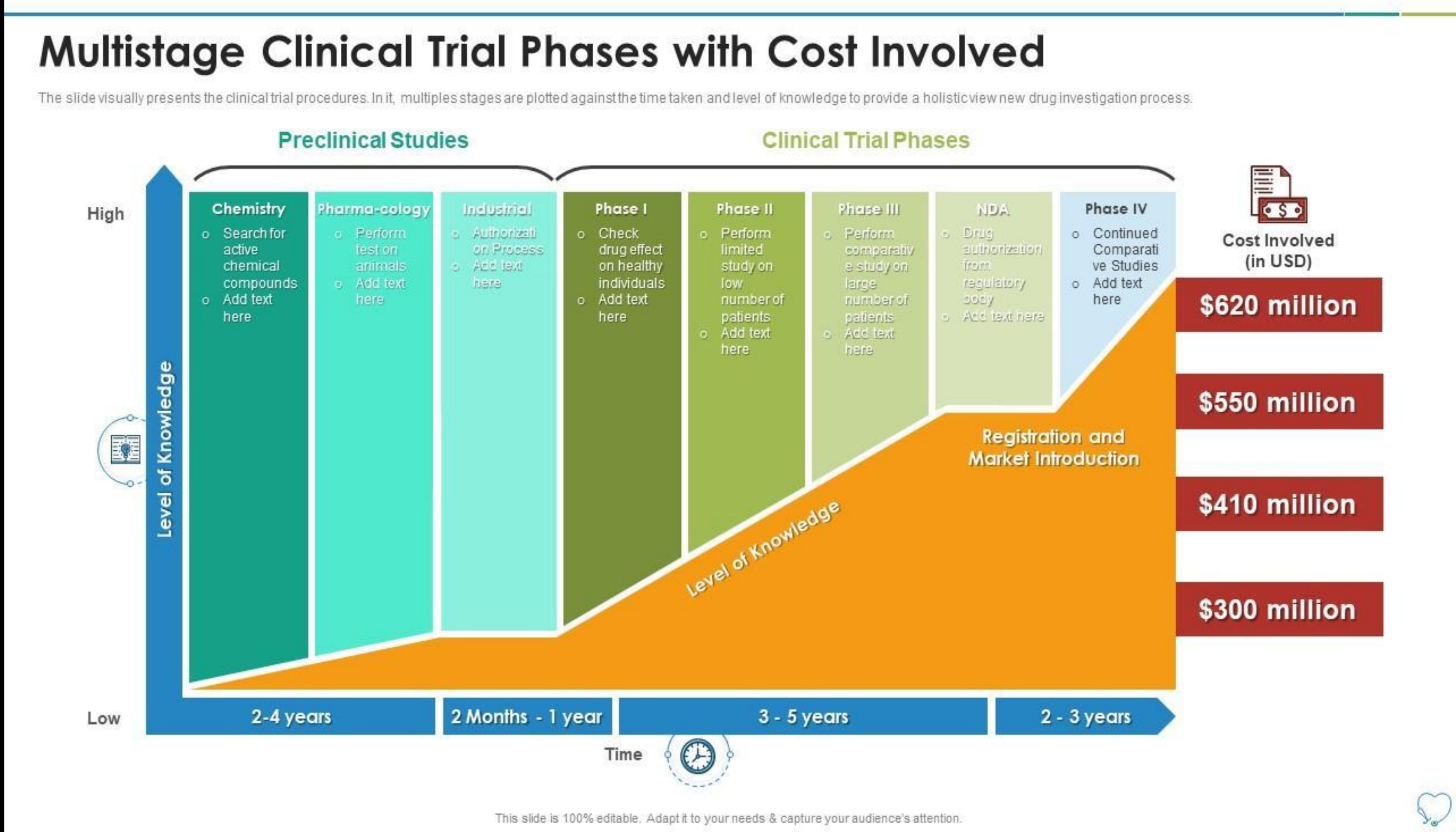
INTRODUCTION: INNERWORKINGS AND ASSET ANALYSIS

- Drugs go through clinical phases to reach market approval and gain cash flows
- Lack of knowledge in the sector from the general investor leaves a knowledge gap
- This knowledge gap in asset analysis leads to high volatility market displacement into easier to value assets



INTRODUCTION: INNERWORKINGS AND ASSET ANALYSIS

- When inflation rises micro-small cap only experience rise in clinical costs
- Other companies can raise prices along with
- Micro and small cap biotech need cash runways to gain market approval
- Cash runways are hurt by incorrect equity evaluations
- Can be directly caused by macroeconomic conditions and investor sentiment thereof
- This has caused a “biotech desert” in recent years (2021-2025)



LITERATURE REVIEWS

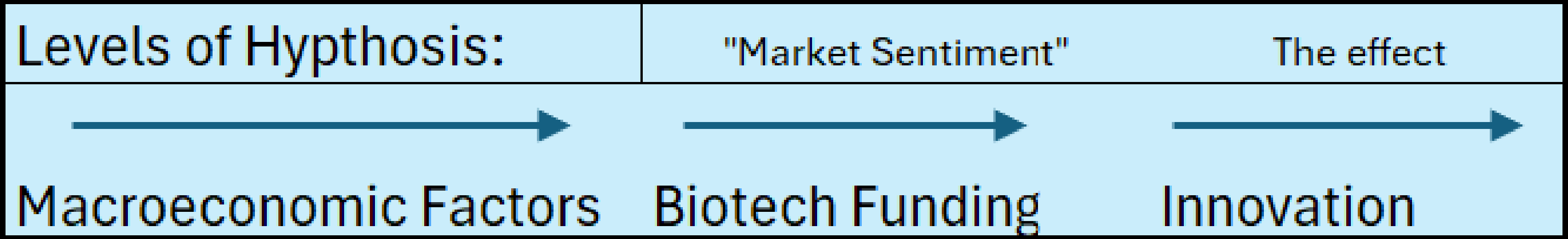
- Title: " Inflation and Innovation Value: How Inflation Affects Innovation And The Value Strategy Across Firms."
- Authors: Leonardo Andrade Rocha and Leonardo Querido Caceres
- "Rising inflation promotes a reduction in R&D investments demand, reducing the rate of technological process."
- This supports the thesis that periods of increased inflation then leads to periods of higher interest rates to counteract.
- Firms are aware of this and are risk averse during times of rising inflation.
- Source: https://www.redalyc.org/journal/5723/572365672006/html/

- Title: "Financial Intermediation and the Funding of Biomedical Innovation: A Review"
- Authors: Andrew W. Lo and Richard T. Thakor
- How funding for Biotech companies is a nuanced endeavor
- The niche analysis needed in this sector raises the need for Financial Intermediaries.
- Need for Concentrated investments raises need for Financial derivatives
- Source: https://carlsonschool.umn.edu/sites/carlsonschool.umn.edu/files/2022-04/JFI%20Review%20Lo%20Thakor%20v5.pdf

- Title: "Biotech Companies Challenged in Times of Inflation"
- Author: Peter Thilo Hasler
- Covers how inflation in specific creates a gap between companies with already established drugs ("Flight to Quality")
- Companies that have drugs on the market can raise the price of said drugs to surpass inflation.
- Companies that are in preclinical development face higher costs in R&D and thus have inverse relation to the rising of inflation.
- In addition, this relative unattractiveness in business performance for preclinical companies lowers the probability of receiving financing directly.
- Source: https://european-biotechnology.com/opinion/biotech-companies-challenged-in-times-of-inflation/

RESEARCH QUESTION AND HYPOTHESIS

- **<u>Research Question:</u>**
- How do macroeconomic conditions and financial market cycles, monetary short-termism, and cyclical emotionalism thereof effect innovation through investment activity- especially as it pertains to micro and small cap firms?
- **<u>Hypothesis:</u>**
- Macroeconomic factors-such as interest rates, inflation, monetary supply and investor sentiment thereof will hold innovation hostage or help it
- Clinical Biotech are especially sensitive yet are especially necessary for innovation



BASE REGRESSION VARIABLES AND EMPIRICAL APPROACH

- **Dependent Variables:**
- NBI (NASDAQ Biotech Index)- represents biotech firms with established revenues
- BBC (Virtue LifeSci Biotech Clinical Trials ETF)- Represents clinical stage biotech firms
- **Independent Variables:**
- Monetary factors: 10 yr yield, Money supply (M2)
- Market Conditions: Break even Inflation Rate (BEIR), VIX (Volatility "fear" index)
- Credit Market: ICE BofA BBB US Corporate Index- lower quality corporate credit conditions (borrowing rate for micro-small cap biotech)
- **Controls:**
- S&P 500- understood large cap market index
- NASDAQ Composite- understood technology index

SUMMARY STATISTICS OF BASE REGRESSION (NO CONTROLS)

BBC (Life Science Clinical stage Index phase 1-3)		*NBI (Nasdaq Biotechnology Index)*		*BEIR (2003-2025), Inflation Rate (1993-2003)*		*Corporate Index Total Return Index Value*		*at 10-Year Constant Maturity, Quoted on an Investment Basis*		*CBOE Volatility Index: VIX*		*M2*	
Mean	29.105	Mean	1826.725	Mean	2.093006	Mean	543.3919	Mean	3.865623	Mean	19.74257	Mean	9997.544
Standard Error	0.769885	Standard Error	79.13179	Standard Error	0.033344	Standard Error	13.6273	Standard Error	0.086481	Standard Error	0.397622	Standard Error	293.8221
Median	27.26346	Median	948.6709	Median	2.1	Median	474.1	Median	3.9	Median	17.82	Median	8462.55
Mode	#N/A	Mode	#N/A	Mode	2.7	Mode	261.86	Mode	6.27	Mode	12.47	Mode	#N/A
Standard Deviation	8.607576	Standard Deviation	1536.461	Standard Deviation	0.647416	Standard Deviation	264.5943	Standard Deviation	1.679152	Standard Deviation	7.720421	Standard Deviation	5697.423
Sample Variance	74.09037	Sample Variance	2360714	Sample Variance	0.419148	Sample Variance	70010.14	Sample Variance	2.819552	Sample Variance	59.60489	Sample Variance	32460627
Kurtosis	0.863359	Kurtosis	-1.09486	Kurtosis	2.421153	Kurtosis	-1.35078	Kurtosis	-0.75875	Kurtosis	6.482431	Kurtosis	-0.54485
Skewness	1.120204	Skewness	0.677708	Skewness	-0.44824	Skewness	0.25094	Skewness	0.231294	Skewness	1.978409	Skewness	0.794895
Range	40.24379	Range	5169.839	Range	5.198889	Range	848.88	Range	7.33	Range	52.54	Range	18288
Minimum	16.80362	Minimum	147.725	Minimum	-1.39889	Minimum	170.3	Minimum	0.63	Minimum	10.13	Minimum	3474.5
Maximum	57.04741	Maximum	5317.564	Maximum	3.8	Maximum	1019.18	Maximum	7.96	Maximum	62.67	Maximum	21762.5
Sum	3638.125	Sum	688675.5	Sum	789.0632	Sum	204858.7	Sum	1457.34	Sum	7442.95	Sum	3759077
Count	125	Count	377	Count	377	Count	377	Count	377	Count	377	Count	376

- BBC (25.10) and NBI (1826.72) mean value and size difference
- BBC shows greater relative volatility with a larger relative standard deviation percentage compared to the NBI
- Inflation Rate (BEIR): short spikes choke clinical funding, raw rates were high, but expectation remained low surprisingly
- Market Volatility (VIX): Several heightened vol periods
- ICE BofA BBB: Substantial variability: - reflects changing credit conditions-. Treasury yields averaged 3.87% with a standard deviation of 1.67%, representing the variable interest rate environment that biotech companies faced during the study period

CONTROL SUMMARY STATISTICS

- More companies in the NASDAQ
- Highest Market cap companies from the NASDAQ and overall market make up the S&P 500
- S&P has been known to represent overall market sentiment
- Though it really just measures sentiment of the largest companies in the world
- So why is monetary policy tailored towards this index?

NASDAQ Composite Index		*S&P 500*	
Mean	4818.198	Mean	160.17
Standard Error	233.6582	Standard Error	6.980827
Median	2655.076	Median	96.15522
Mode	#N/A	Mode	#N/A
Standard Deviation	4536.822	Standard Deviation	135.5431
Sample Variance	20582752	Sample Variance	18371.94
Kurtosis	1.378396	Kurtosis	1.274225
Skewness	1.541009	Skewness	1.466652
Range	19021.07	Range	574.8245
Minimum	713.49	Minimum	25.6728
Maximum	19734.56	Maximum	600.4973
Sum	1816461	Sum	60384.08
Count	377	Count	377

BASE CORRELATION MATRIX (W/CONTROLS)

	Inflation	*BBB*	*10 Yr*	*VIX*	*M2*	*NASDAQ*	*S&P 500*
BEIR (2003-2025), Inflation Rate (1993-2003)	1						
ICE BofA BBB US Corporate Index Total Return Index Value	-0.16281	1					
Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity, Quoted on an Investment Basis	0.028471	-0.41024	1				
CBOE Volatility Index: VIX	0.149624	-0.37454	-0.22903	1			
M2	0.087986	-0.02945	-0.25107	0.160493	1		
NASDAQ Composite Index	-0.15359	0.3559	0.154274	-0.58817	-0.07398	1	
S&P 500	-0.22884	0.462461	0.169358	-0.71201	-0.10579	0.866483	1

- **Credit, Inflation, and Volatillity are interconnected:** This supports hypothesis that Inflation and volatility create a harder to borrow environment and effect biotech equity valuations- negatively correlated with both

- **The Nasdaq Composite and S&P 500** show a strong positive correlation (.866), though they exhibit different relationships with other variables, potentially explaining their divergent effects in the regression models.

- **Market volatility (VIX)** shows negative correlations with most market indices, thus confirming it role as a counter cyclical indicator and "fear" as it pertains to overall market sentiment.

BBC (NO CONTROL)

- m/m rates
- BBB strong correlation with BBC confirming corporate bond dependency coefficient= 2.456, t-stat = 3.96, p< .001
- Treasury Yields demonstrate a significant positive relationship with BBC (coefficient=.234, t-stat=2.32, p=.022)
- Money Supply (M2) displays a significant positive relationship with the BBC (coefficient = 2.058, t-stat= 2.18, p=.031), indicating that monetary policy conditions substantially influence clinical-stage biotech companies.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.511536
R Square	0.261669
Adjusted R Square	0.230383
Standard Error	0.078551
Observations	124

ANOVA

	df	*SS*	*MS*	*F*	*ignificance F*
Regression	5	0.258037409	0.051607	8.363967	8.51E-07
Residual	118	0.728085512	0.00617		
Total	123	0.986122921			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*.ower 95.0%*	*Jpper 95.0%*
Intercept	-0.01803	0.00920975	-1.95814	0.052574	-0.03627	0.000204	-0.03627	0.000204
BEIR (2003-2025), Inflation Rate (1993-2003)	0.025165	0.080451169	0.312796	0.754988	-0.13415	0.18448	-0.13415	0.18448
ICE BofA BBB US Corporate Index Total Return Index Value	2.455958	0.620444465	3.958386	0.000129	1.227309	3.684607	1.227309	3.684607
Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity, Quoted on an Investment Basis	0.234311	0.101026238	2.319312	0.022098	0.034252	0.434371	0.034252	0.434371
CBOE Volatility Index: VIX	-0.03667	0.037283156	-0.98356	0.327346	-0.1105	0.037161	-0.1105	0.037161
M2	2.058448	0.942202752	2.184718	0.030886	0.19263	3.924266	0.19263	3.924266

BBC (W/ CONTROL)

- m/m rates
- The baseline relationships remain consistent
- Using NASDAQ and S&P 500 we can control overall investment activity within the market
- negative correlation with the S&P speaks to the volumes of potential economic uncertainty being attributable to opportunistic costs in the sector
- creating an inefficient blow off the top market through flight to quality

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.701232673							
R Square	0.491727262							
Adjusted R Square	0.460788921							
Standard Error	0.065932019							
Observations	123							
ANOVA								
	df	*SS*	*MS*	*F*	*ignificance F*			
Regression	7	0.483635378	0.06909	15.8938	1.8E-14			
Residual	115	0.499908581	0.00435					
Total	122	0.98354396						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*Jpper 95.0%*
Intercept	-0.022506721	0.007991173	-2.81645	0.00572	-0.03834	-0.00668	-0.03834	-0.00668
BEIR (2003-2025), Inflation	0.009464211	0.068286729	0.1386	0.89001	-0.1258	0.14473	-0.1258	0.14473
ICE BofA BBB US Corporate	1.2751776	0.629109549	2.02696	0.04498	0.02903	2.52132	0.02903	2.52132
Market Yield on U.S. Treasur	0.207993582	0.09264348	2.2451	0.02667	0.02448	0.3915	0.02448	0.3915
CBOE Volatility Index: VIX	0.034846785	0.03908282	0.89161	0.37446	-0.04257	0.11226	-0.04257	0.11226
M2	0.85039132	0.818844474	1.03853	0.3012	-0.77158	2.47236	-0.77158	2.47236
NASDAQ Composite Index	2.224477611	0.414095912	5.37189	4.1E-07	1.40423	3.04472	1.40423	3.04472
S&P 500	-1.359921895	0.642366123	-2.11705	0.03641	-2.63233	-0.08752	-2.63233	-0.08752

PROOF OF CONCEPT: SUMMARY STATISTICS

- Regression is now y/y
- NME=New molecular entity
- Direct sign of innovation
- Lag as clinical timelines are around 10 yrs
- New molecular entity stands for a mechanism of action that has not yet been approved
- This means that a new drug is approved for market consumption
- A drug that could drastically change healthcare industry and improve/save someone's life
- This would be a direct indicator of "buying time"

NME	
Mean	33.35484
Standard Error	2.098924
Median	30
Mode	22
Standard Deviation	11.68631
Sample Variance	136.5699
Kurtosis	-0.74873
Skewness	0.518368
Range	42
Minimum	17
Maximum	59
Sum	1034
Count	31

PROOF OF CONCEPT

- The statistically significant negative coefficient indicates that stronger S&P performance predicts fewer new drug approvals in subsequent periods
- Suggests that capital concentration in large-cap stocks (as seen during the 2021-2025 "biotech desert")
- With other regression intuition
- High inflation and Volatility not only hurts equity valuation but causes "flight to quality in S&P"
- This supports the butterfly effect hypothesis

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.399336
R Square	0.159469
Adjusted R	0.112773
Standard E	0.34292
Observatio	20

ANOVA

	df	*SS*	*MS*	*F*	*ignificance F*
Regression	1	0.401588	0.401588	3.415034	0.081105
Residual	18	2.116697	0.117594		
Total	19	2.518285			

	Coefficients	*tandard Errc*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*Jpper 95.0%*
Intercept	0.126556	0.086309	1.466315	0.159815	-0.05477	0.307883	-0.05477	0.307883
SPY	-0.75988	0.411193	-1.84798	0.081105	-1.62376	0.104008	-1.62376	0.104008

TAKEAWAYS

- Financial intermediaries bridge knowledge gaps in clinical asset analysis and allow equities to have accurate valuations that help innovation
- Financial derivatives should be implemented to help concentrated investments and allow for less of a knowledge gap
- Monetary policy should consider:
- highly volatile moves
- highly inflationary moves
- safeguarding of the BBB corporate bond index
- This is all under the assumption that we would like humanity to live longer and higher quality lives of course…

LIMITATIONS

- Will have to read paper to see other regressions and scenario regressions
- I will get into more of why I used the NASDAQ in the paper
- Can build more intuition and have more effect in the paper
- The error term represents what is not explainable through FDA regulation changes
- Would like to show you my other regression but due to time I will cut it here

DATA SOURCES

- https://www.investing.com/etfs/spdr-s-p-biotech-chart
- https://indexes.nasdaqomx.com/Index/History/NBI
- https://www.macrotrends.net/1320/nasdaq-historical-chart
- https://www.forbes.com/advisor/investing/fed-funds-rate-history/
- https://www.ssb.no/en/priser-og-prisindekser/konsumpriser/statistikk/konsumprisindeksen
- https://www.statista.com/statistics/716586/total-biotechnology-funding-by-the-national-institutes-for-health/
- https://www.fda.gov/about-fda/histories-fda-regulated-products/summary-nda-approvals-receipts-1938-present
- http://www.mtspartners.com/wp-content/uploads/sites/2/2016/08/IPO-Monitor-12.07.17_Redacted-1.pdf
- https://medical-technology.nridigital.com/medical_technology_jan25/investors_optimistic_about_continued_biotech_and_medtech_ipo_resurgence_in_2025
- https://godelterminal.com/
- https://www.bamsec.com/
- https://seekingalpha.com/
- https://lifesciencereport.com/
- https://bowtiedbiotech.com/
- https://fred.stlouisfed.org/
- Pandas" data reader in Python yfinance
- FRED ST. Louis